

RESEARCH BRIEF · METR 2025

# Measuring AI Ability to Complete Long Software Tasks

*Time Horizon Doubling Every 7 Months*



month doubling time  
50% task-completion horizon

Thomas Kwa, Ben West, Joel Becker, et al. · Model Evaluation & Threat Research (METR)

[arxiv.org/abs/2503.14499](https://arxiv.org/abs/2503.14499)

# A New Metric: 50% Task-Completion Time Horizon

1

## Assemble Task Suite

Collect 170 diverse software & research tasks ranging from 3 sec to 8 hrs of human effort.

2

## Collect Human Baselines

Measure how long skilled professionals take to complete each task; 800+ baselines across 2,529 hours.

3

## Fit Logistic Model

Test AI agents on tasks; plot success probability vs. task duration. Find the duration where AI succeeds 50% of the time.

| Family             | Length     | Description                                                                                                                                                               |
|--------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| find_shell_script  | 3 seconds  | Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"                                                                         |
| wikipedia_research | 1 minute   | Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.                                                             |
| oxdna_simple       | 9 minutes  | Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.                                                                      |
| munge_data         | 56 minutes | Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.                                    |
| cuda_backtesting   | 8 hours    | Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement. |

Fig 1. Methodology overview: 170 tasks → human timing → AI success curve

# 170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

## Benchmark Composition

| Source       | Tasks      | Time Range     |
|--------------|------------|----------------|
| HCAST        | 97         | 1 min – 30 hrs |
| RE-Bench     | 7          | 8 hrs (all)    |
| SWAA         | 66         | 1 sec – 30 sec |
| <b>TOTAL</b> | <b>170</b> | 3 sec – 8 hrs  |

## Example Tasks & Durations

| Task               | Duration | Description                            |
|--------------------|----------|----------------------------------------|
| find_shell_script  | 3 sec    | Which file is a shell script?          |
| wikipedia_research | 1 min    | Research factual info on Wikipedia     |
| oxdna_simple       | 9 min    | Debug molecular dynamics input files   |
| munge_data         | 56 min   | Write Python script to transform JSON  |
| cuda_backtesting   | 8 hrs    | Implement CUDA kernels for 30× speedup |

**2,529**

Human Hours  
Collected

**800+**

Human  
Baselines

**3 sources**

HCAST · RE-Bench  
· SWAA

**6 yrs**

Model Release  
Span

# AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years

## Key Findings

**~207days**

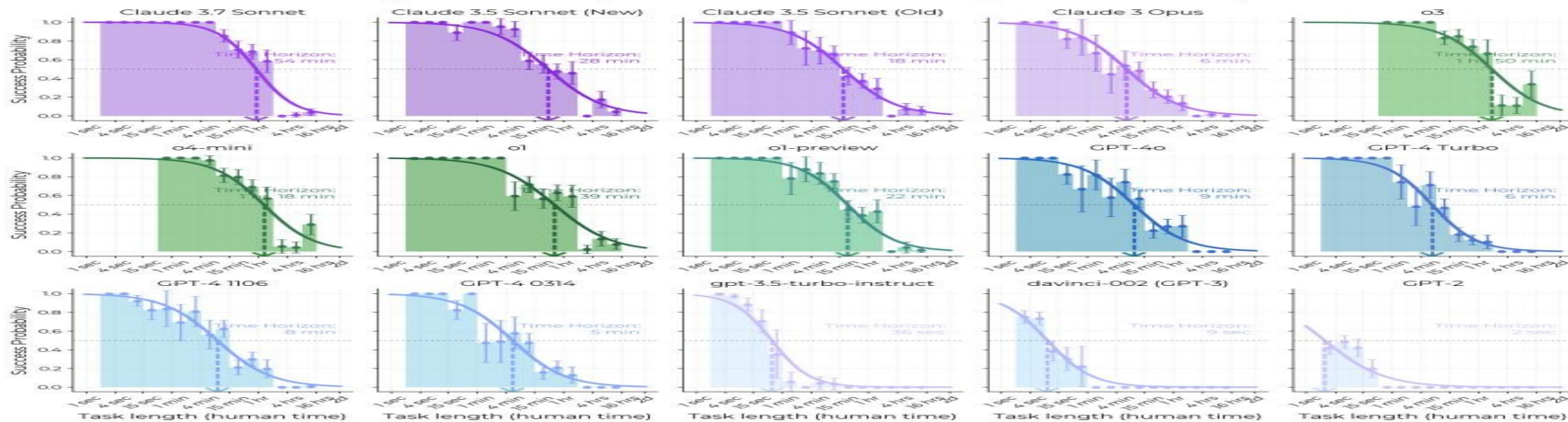
Doubling Time (~7 months)

**$R^2=0.97$**

Exponential fit quality

**6×**

Time horizon growth  
GPT-4 → o3



# Success Probability Drops Sharply as Task Length Increases

## What the Curves Show

- Each curve = one model
- X-axis: task duration (log scale)
- Y-axis: success probability
- Curves shift right over time
- Frontier models (o3) sustain higher success at longer tasks
- Claude 3.7 Sonnet (54 min) vs. GPT-2 (2 sec) — same probability at vastly different task lengths

The shift to the right means frontier models handle complex, multi-step tasks that earlier models could not attempt.

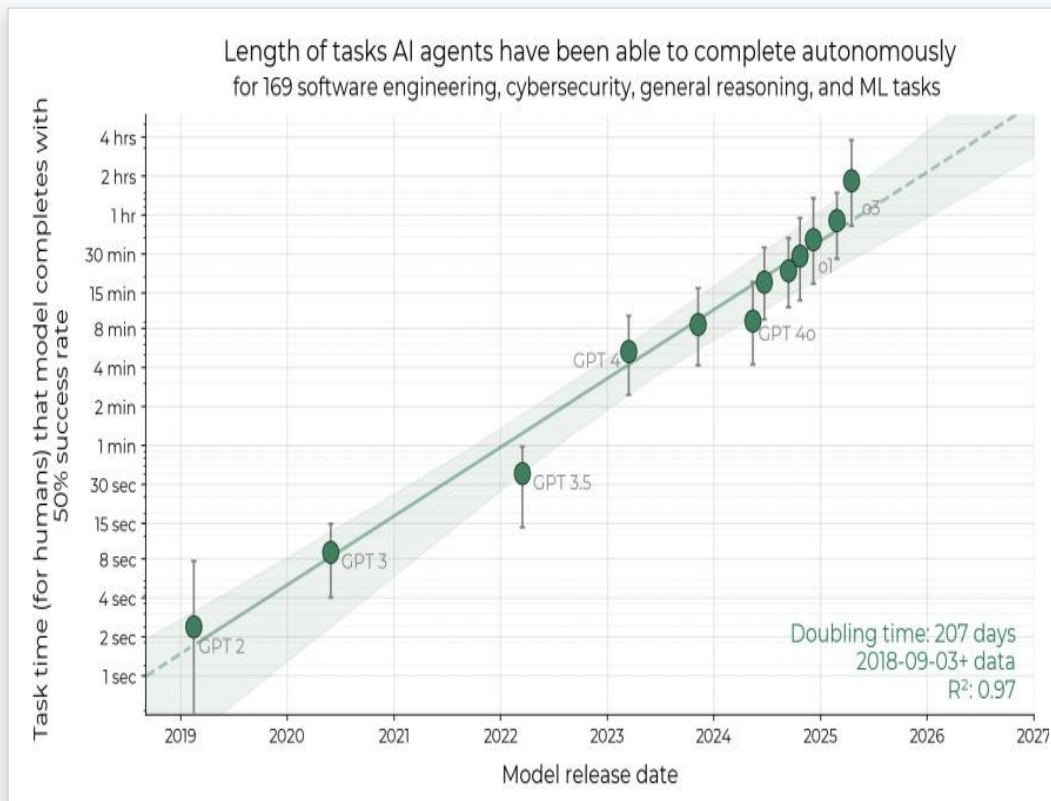


Fig 3. Per-model success probability vs. task duration — frontier models extend the feasible task frontier rightward

# Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

## → Improved Logical Reasoning

Chain-of-thought and extended reasoning chains allow models to plan multi-step solutions rather than just executing single operations.

## Better Tool Use

Models now reliably execute shell commands, call APIs, read/write files, and orchestrate multiple tools within a single task session.

## ⦿ Greater Reliability & Self-Awareness

Self-correction, uncertainty acknowledgment, and know-when-to-stop behavior reduce failure modes on long-horizon tasks.

## ↺ Error Adaptation

Frontier models detect failed steps, diagnose root causes, and recover — rather than dead-ending after a single error.

**Limitation:** Performance is notably lower on less structured, "messier" tasks — real-world deployments may face harder distributions than the benchmark suite.

# Extrapolation Predicts Month-Long Task Automation by 2028–2031

## → 1-month horizon (167 work hours)

Naive extrapolation of the 2019–2025 exponential trend predicts frontier AI will reach this milestone between:

**mid-2028 — mid-2031**

### ~20% Faster Since 2023

The 2023–2025 growth rate is approximately 20% faster than the overall 2019–2025 rate.

This suggests possible acceleration — not slowing — in the time-horizon expansion.

### 80% Horizon ~5× Shorter

Tasks completed 80% of the time show a similar exponential trend, with horizons roughly 5× shorter than the 50% horizon.

For o3: 50% at ~2 hrs; 80% at ~25 min.

### Caveats

- External validity remains uncertain — benchmark tasks may not perfectly represent real-world intellectual labor
- Trend changes (regulation, capability plateau, compute constraints) could break the extrapolation
- Supplementary experiments find little evidence trends are slower on more realistic tasks, but sample is limited

# Takeaways

1 The 50% time horizon is a robust, intuitive metric —  $R^2 = 0.97$  over 6 years of model releases.

2 Doubling every ~7 months, with possible acceleration since 2024 — capability growth is not slowing.

3 Frontier models (o3) autonomously complete ~2 hours of human-equivalent software work — up from 2 seconds for GPT-2 in 2019.

4 External validity remains open — benchmark tasks may not fully represent average real-world intellectual labor distributions.

5 Key capability drivers: reasoning, tool use, error recovery. Key limitations: unstructured tasks, uncertain generalization.

*Exponential autonomous capability growth demands proactive safety governance — not reactive oversight.*